

Debiasing Estimates of Global Forest Cover Loss

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Expanded access to satellite data, reduced computational costs, and continued advances in machine learning have ushered in a measurement revolution across the social, economic, and environmental sciences. Innovations in remote sensing now allow for the quantification of previously difficult-to-measure outcomes of interest at high spatial resolution over time. These data enable more precise evaluation of the causal impacts of policies, which is crucial to the design and implementation of interventions that effectively combat climate change, alleviate poverty, and improve food security.

A common approach is to train a machine learning model to predict some outcome Y using satellite features k , generating predictions $\hat{Y}$. However, when these models are trained by choosing parameters to minimize a standard loss function, the resulting predictions can produce biased estimates when estimating regression coefficients. If the measurement error in the outcome variable is correlated with policy variables or important confounders, as is the case for many widely used remote sensing data sets, estimates of the causal impacts of interventions will be biased. This bias can occur even in cases when researchers have a good instrument or an experimental research design.

New methods have been developed to correct these biases, however, most methods require some representative ground-truth labeled data. In this paper, we use active learning techniques to collect ground truth data on deforestation in Africa and check for biases in widely used satellite-derived data on forest cover change.

I. Prediction Errors and Causal Inference

To fix ideas, consider a regression with one independent variable X measured for observations indexed by i . We seek to estimate the effect of changes in X and an outcome Y according to:

$$(1) \quad Y_i = \alpha + \beta X_i + e_i.$$

In this case, β measures the marginal effect of X on Y . However we do not have access to the true Y_i for all observations. Instead we have predictions from a machine learning model: $\hat{Y}_i = Y_i + v_i$, where v_i is the measurement error for a given observation.

When we estimate β from (1) using OLS with $\hat{Y}_i$ as the dependent variable instead of Y_i , our estimate of β will equal in expectation:

$$(2) \quad \mathbb{E}[\hat{\beta}] = \beta + \frac{\text{Cov}(e, X)}{\text{Var}(X)} + \frac{\text{Cov}(v, X)}{\text{Var}(X)}.$$

The middle term in the equation is standard endogeneity bias. The third term is the result of using machine learned proxies for Y_i instead of the true values. In words, the measurement error from our machine learning model will bias our estimate if it is correlated with the treatment variable. This can happen even when X_i is assigned randomly. For example, if a treatment has no effect on outcomes of interest, but it induces non-random measurement error, researchers could measure a spurious effect where none existed (see panel (d) in Figure 1).

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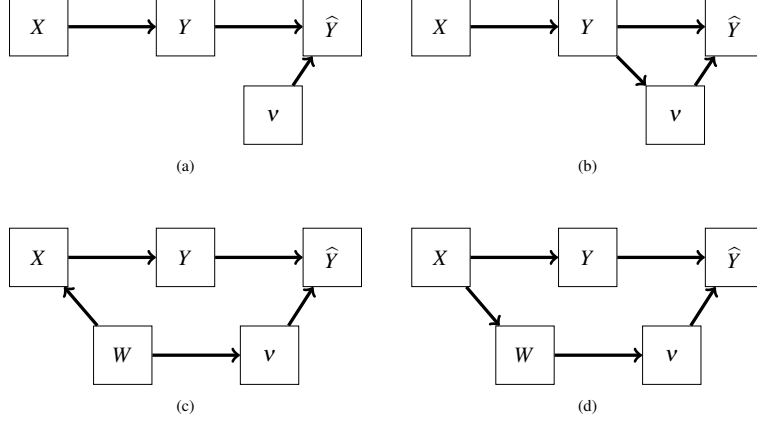


Figure 1. : Four Directed-Acyclic-Graphs

Note: Diagrams illustrate potential relationships between treatment (X), outcomes (Y), measurement error (v), machine learning model predictions ($\hat{Y}$), and unobserved variables (W). (a) shows classical measurement error, (b) shows outcome-induced bias, (c) shows confounder-induced bias, and (d) shows treatment-induced bias.

A. Related Work

New work on this topic shows how to correct for this type of error using a subsample of points where the ground truth measures of Y are known. One approach, prediction-powered-inference, develops a loss function for the downstream estimator that corrects for biases (Angelopoulos et al., 2023; Carlson and Dell, 2025). A related method known as predict-then-debias, estimates the bias and subtracts it from the coefficient of interest (Kluger, 2025). A third set of papers use a variety of approaches to correct the predictions themselves (Ratlidge, 2021; Proctor, Carleton and Sum, 2023; Battaglia et al., 2024; Rambachan, Singh and Viviano, 2025).

In Sanford et al. (2025), the authors investigate prediction models that avoid differential measurement error by construction. We want a predictive model with parameters ω^* that satisfies:

$$(3) \quad \omega^* = \arg \min_{\omega} L_p(\hat{Y}(\omega), Y, k) \quad \text{s.t.} \quad \text{Cov}(X, Y - \hat{Y}(\omega)) = 0$$

where L_p is a loss function, such as mean squared error. In that paper, the authors show that the adversarial debiasing framework proposed by Zhang, Lemoine and Mitchell (2018), which was originally used to debias machine learning model predictions with respect to race or gender, can be isomorphic to the constrained loss function above under certain circumstances. Roughly speaking, in an adversarial debiasing setup, a primary model simultaneously tries to minimize its own loss function, while maximizing the loss function of an adversarial model.

We show that, if the adversary is a linear regression of the form:

$$(4) \quad v_i = \gamma X_i + \varepsilon_i,$$

then the primary model will try to make predictions such that the errors are uncorrelated with the treatment variable, exactly the condition needed to ensure unbiased estimates of β in equation 1.

We show that this method works well in a number of simulations. As with many of the other methodologies that can be used to address prediction error biases, however, a major obstacle to implementing this method is a lack of ground truth data needed to debias predictions.

II. Case Study: Hansen Global Forest Change

The Global Forest Change (GFC) data estimates forest cover loss globally at high resolution using a machine learning model trained on satellite imagery Hansen (2013). These estimates are widely used to study the drivers of deforestation. They have also been found to contain non-classical measurement error in certain settings. Tropek (2014) showed that the data often confuses tree plantations for forests, which may be relevant for research questions that study deforestation. Bastin (2017) showed significant underestimation of forest cover in a cross-section of ground truth points in global dryland regions.

For most applications in social sciences that attempt to answer causal questions, we would need ground truth data on changes in deforestation rates over time to assess biases. One potential source of such data is Guo, Zhu and Gong (2022) (hereafter, Guo). The authors select a set of pixels and visually interpret high-resolution historical imagery from various sources to manually label forest cover change from 2000-2020. In the left panel of Figure 2, we match these pixels to the corresponding GFC predictions¹ and calculate v_i , treating the Guo labels as the ground-truth. We count a prediction of deforestation as accurate if the pixel was deforested in any year, not necessarily the exact year that was predicted. We then aggregate the average prediction error to the country level, attempting to account for their unequal probability sampling strategy. Positive values (countries in green) indicate that the predictions tend to overestimate deforestation relative to the Guo labels, while negative values (countries in red) indicate that the predictions underestimate deforestation.

While sample sizes for some countries are small, the map shows that the GFC predictions may be differentially biased across countries. There are several issues with the Guo labels that limit their usefulness, however. The authors do not record their probabilities of sampling within each strata, and furthermore, they drop some observations, making it difficult to determine representativeness. Furthermore, the sample is small and mostly static. For example, there are only 149 observations in Africa that show forest cover loss, making it difficult to have enough power to detect potential biases. For these reasons, we undertake our own data collection exercise in Africa to build a more complete and representative labeled data set on the continent with the highest rates of deforestation.

A. Active Sampling and Approximately Optimal Verification of Forest Change in Africa

To collect this data, we use approximately optimal sampling methods proposed by Zrnic and Cands (2024) and Gordon and Papp (Forthcoming). The basic idea is to first train a machine learning model on whatever non-representative ground-truth data is available. In our case, we use the Guo data, and train a 3-layer LSTM neural network (Hochreiter and Schmidhuber, 1997).

Next we use this model, which, given the minimal training data, has fairly low out-of-sample-accuracy and is likely biased, to make a preliminary set of predictions in a representative sample of unlabeled points. We use these predictions in combination with Neyman (1934) optimal sampling methods to choose a representative set of these points to label, sampling points with probability $\pi_i \propto \sqrt{p_i(1-p_i)}$, where p_i is our model's predicted probability of deforestation for pixel i . This sampling strategy leads us to over sample points that the model is most uncertain about. The usefulness of this procedure comes from the fact that, even if the p_i are miscalibrated or biased, we will still end up with a representative sample (once labeled observations are weighted by $1/\pi_j$). Any biases in the preliminary model thus only affect the efficiency of our resulting estimates, not the bias.

We sample 2,000 pixels using this method. We then verify whether deforestation has occurred in these pixels using visual interpretation of historical high-resolution satellite imagery using Google Earth Pro. For each point, we have two annotators independently inspect all available imagery, and mark the first year between 2000-2017 in which deforestation is observed. If the annotators disagree, we have one of the authors break the tie. To mimic the structure of the GFC data, we define a forested pixel as a having 30% tree cover (vegetation > 5m tall), and deforestation as the loss of more than half

¹2023 Version 1.11.

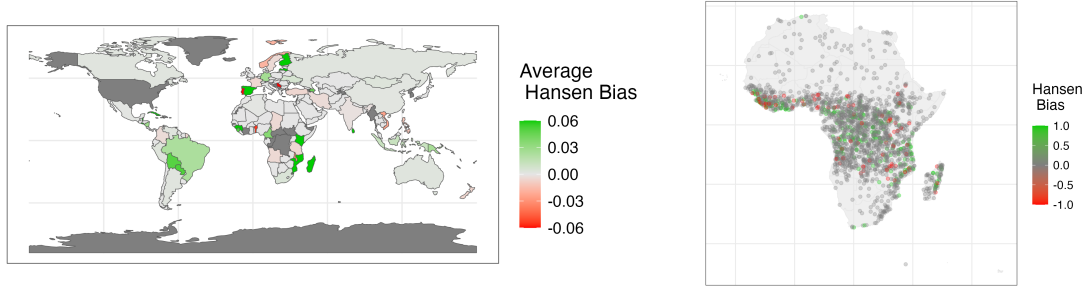


Figure 2. : Measurement Error in Global Tree Cover Loss Estimates

Note: Left panel shows average bias in GFC predictions calculated using Guo, Zhu and Gong (2022) ground truth data. Values beyond ± 0.06 saturated. Right panel shows bias for a sample of pixels in Africa selected using active learning techniques. Positive values indicate predicted deforestation where none occurred in any year (false positives), and negative values indicate deforestation occurred where it was not predicted (false negatives).

of pre-existing tree-cover. As with their data, we do not record multiple instances of deforestation in the same location or regrowth. This process is imperfect, especially in the earlier years of our study period when high-resolution data is less frequently available, yet we believe it offers a substantial improvement over current alternatives, and it should prove valuable to researchers in stimulating further investigation when bias is detected.

Figure 2 shows a map of our labeled points. Note that the active sampling strategy leads us to over sample in the forested regions near the equator, which is where the most deforestation occurs. In contrast, we sample relatively fewer points in desert regions, where our model was fairly certain that no deforestation had occurred. The points are colored by the GFC prediction errors — there appear to be geographic trends. Green points are false positives, points predicted to have deforestation that were not deforested. These appear to be more concentrated in the most densely forested regions of central Africa. Red points show false negatives — points that saw deforestation but were missed by the GFC predictions. These appear to be more common in areas with lower baseline forest cover.

III. Results and Conclusion

To test for geographic biases, we run the following simple regression:

$$(5) \quad \hat{Y}_i = \beta_1 Y_i + \beta_2 \bar{Y}_i^n + e_i,$$

where $\hat{Y}_i$ is a dummy for GFC predicted deforestation at pixel i in grid cell g in any year, Y_i is true deforestation at that pixel in any year, and $\bar{Y}_i^n$ is average observed deforestation in the 10 nearest neighbor labeled pixels, calculated using the sample weights. Results in Appendix Table A1 show a positive and significant estimate for β_2 , indicating that nearby deforestation positively correlated with GFC predicted deforestation, even after controlling for whether the pixel was actually deforested. These results show that the GFC data predicts relatively more deforestation in the areas where deforestation occurs, indicating spatial clustering of prediction errors.

These results demonstrate that empirical work that attempts to infer causal effects of policies using machine-learned variables needs to carefully consider how differential prediction error will affect the analysis. Recent research in this area offers a variety of methods to detect and correct for potential biases, however most of these methods rely on a representative sample of ground-truth data. In this paper, we collect ground truth data on deforestation in Africa and assess commonly used satellite-derived measures of deforestation for bias. There is a great need for future research in this area to

determine which methods work best under which circumstances, determine more efficient ways of collecting ground-truth data, and gathering ground-truth data for a greater variety of outcomes.

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APPENDIX

	<i>GFC Predicted Deforestation</i>			
	(1)	(2)	(3)	(4)
(Intercept)	0.010*** (0.003)	0.008* (0.003)	−4.591*** (0.044)	−4.719*** (0.047)
Deforested	0.232*** (0.016)	0.226*** (0.016)	3.449*** (0.073)	3.248*** (0.076)
Neighbor Mean Deforested		0.065* (0.028)		2.444*** (0.241)
Num.Obs.	2000	2000	2000	2000
Model	OLS	OLS	Logit	Logit

Table A1—: Predicted Deforestation

Results from estimating equation 5. Dependent variable is an indicator for GFC predicted deforestation (Hansen, 2013). Deforested is an indicator for whether the pixel was labeled as deforested in high resolution satellite imagery, and in columns 2 and 4, the average deforestation labels for the 10 nearest neighbor pixels. Columns 1 and 2 estimate a linear probability model with OLS, columns 3 and 4 estimate a logit model. Observations are weighted by the inverse probability of sample inclusion.